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DI PADOVA

Presentations using the Unipd theme

Demonstrating how to use the Unipd theme

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My implementation of a server/client architecture: concurrent tasks which can be Activated or Disactivated. Focus on :

1. efficiency
2. correct logic
3. effectiveness

Problem Statement: The Real-Time Challenge



- **Objective:** Simulate a system of periodic tasks with strict timing constraints: Execution Time (C_i), Period (P_i), and Deadline (D_i).
- **The Constraint:** In a hard real-time context, adding a new task to a running system is not guaranteed to be safe.
- **Dynamic Environment:** The system must respond to asynchronous client requests to activate or deactivate tasks via a TCP/IP interface.
- **Mathematical Necessity:** Before any task activation, the supervisor must prove that the current task set remains *schedulable* under Rate Monotonic assumptions.